

The ethics of emerging technology

When new tech can do something, should it? This dot point is about the line between can and should — through cyber-security threats, biometric data, and the jobs machines replace.

What this key knowledge covers

This summary distils everything you need for **KK12 — The ethics of emerging technology** in Innovative Solutions. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

PulsePoint — the “frictionless” gym

PulsePoint is a fictional 24/7 gym chain opening near your school. Their pitch is “zero-friction fitness”: no membership cards, no front desk, no queues. You walk up, a camera **scans your face** to unlock the door, a **fingerprint** opens your locker, and an **AI coach app** writes your workout. To make it cheap, PulsePoint has **no reception staff** and only one roaming trainer per three sites.

It’s convenient, modern, and almost everything about it is **legal**. But three quiet questions follow you onto the gym floor — and they’re the exact three this dot point examines:

PulsePoint now holds a database of everyone’s face. What happens when — not if — it gets breached?

Your face and fingerprint are *you*. You can change a password — you can’t change your face.

The receptionist who used to greet you is gone. Where did that job go — and whose problem is that?

Keep PulsePoint in mind. Every concept below maps back to it, and so do the exam answers at the end.

02 What is an “ethical” issue?

An **emerging technology** is one that is new, developing, or being applied in new ways — think AI, facial recognition, drones, wearables, the Internet of Things. Because they’re new, the rules and norms around them often haven’t caught up. That gap is where **ethical issues** live.

A situation that raises questions of **right and wrong** — what a person or organisation *should* do — beyond what the law strictly *requires*. Ethics asks whether an action is fair, honest, and



respectful of others, even when no law is broken.

The line examiners love: legal vs ethical

This is the single most-tested idea in the whole dot point. They are **not the same thing**:

03 Strand A — Cyber-security threats

Every emerging technology that collects data also creates a **target**. The more systems connect (cloud, apps, IoT sensors, smart locks), the bigger the **attack surface** — the number of ways in for an attacker. The *ethical* question isn't "how do firewalls work?" (that's the Cyber Security outcome). It's: **when an organisation collects data, what do they owe the people it belongs to?**

When PulsePoint collects your data, it takes on a moral responsibility to protect it. A weak password policy or an un-patched server isn't just a technical slip — it's a **breach of the trust** users placed in the organisation.

04 Strand B — Biometric data

Biometric data is information about your body or behaviour used to identify you. It splits into two kinds:

Why biometrics are different from a password

This is the heart of the strand, and the best 3-mark answer in any exam. A password is a *secret you choose*. A biometric is a *permanent fact about your body*. That one difference changes everything:

The ethical issues with collecting & storing biometrics

Did the person really agree?

Collected for X, used for Y.

05 Strand C — Job displacement

Job displacement happens when emerging technology — automation, robotics, AI — takes over tasks that people used to be paid to do. PulsePoint's missing receptionist is the everyday face of it; supermarket self-checkouts, automated warehouses, and AI customer-service bots are others.

Displacement = the job disappears. **Deskilling** = the job remains but needs less skill (so it's lower-paid and the skill is lost). **Reskilling / upskilling** = workers learn new skills to move into the jobs technology creates.

It's a balance, not a verdict

Top-band answers refuse the lazy "robots = bad" take. Technology **displaces some jobs and creates others**. The ethical question is about *who wins, who loses, and who is responsible for the people left behind*. Move the automation slider and watch the trade-off:

06 How to analyse any ethical issue

Examiners don't want a rant. They want a **structured analysis**. Use this 4-step frame for every "discuss / analyse the ethical issues" question — it works for all three strands and any new tech they invent on the day.

Decode the command word first

The verb tells you how much to write and what shape your answer takes. Tap a command word to see exactly what the examiner expects.

The P·E·L structure for ethics answers

For 3+ mark ethics questions, build each idea as **Point → Explain → Link**. Make a *point* (name the ethical issue), *explain* the benefit and the harm, then *link* back to the scenario and the people in it. Write an answer below and the trainer checks whether you hit the moves examiners reward.

Six traps that cost marks

Quoting an Act for an ethics question.

Missing the permanence point.

Lists points, never decides.

08 Practice exam questions

Five VCAA-style questions — **17 marks total**. Have a go on paper first, then reveal the weak answer, the strong answer, and the marking guide. Use the self-mark buttons to track your score in the dock.

- 1 — definition: data about a physiological/behavioural characteristic used to identify a person.
- 1 — a valid example (fingerprint, face, iris, voice, gait).
- 1 — a password can be reset/changed after a breach.
- 1 — a biometric is permanent and cannot be changed.
- 1 — consequence: lifelong exposure / reuse for fraud / harm to the member.
- 1 — identifies job displacement as the harm (lost income/role).
- 1 — gives a benefit (lower cost / cheaper access / new roles).
- 1 — names specific stakeholders affected (staff, customers, business).
- 1 — reasoned judgement, e.g. fairness depends on managing the transition (retraining/notice).
- 1 — legal = follows the law; ethical = fairness/right vs wrong.
- 1 — states they are distinct (legal ≠ automatically ethical).
- 1 — applies to PulsePoint (e.g. consent isn't meaningful if you can't refuse).
- 1 — identifies the remaining ethical concern (permanence / misuse / understanding).
- 1 — identifies a genuine benefit of the system.
- 1 — identifies serious ethical costs (permanence / consent / breach target).
- 1 — a clear, justified recommendation that weighs the two.
- 1 — a valid improving condition (opt-in, card alternative, encryption, deletion policy).

Nine things to remember

- **The dot point:** ethical issues of emerging tech — cyber threats, biometric data, job displacement. All three examinable.

- **Ethical ≠ legal.** Legal = what you *can* do; ethical = what you *should* do. The classic trap.
- **Cyber threats:** collecting data creates a *duty of care*; AI, deepfakes and IoT widen the attack surface.
- **Biometrics** = physiological or behavioural traits used to identify you.
- **The crux:** a password can be reset, a biometric cannot — breaches are permanent.
- **Biometric ethics:** consent, permanence, function creep, accuracy/bias, surveillance, storage.
- **Job displacement** ≠ automatically bad — it's about *how* the transition is managed.
- **Always two-sided:** pair a benefit with a harm, name real stakeholders, then conclude.
- **Structure:** Name the issue → stakeholders → weigh perspectives → reasoned position.

Key terms

Term	Meaning
What the law says you can or can't do.	What is right or fair , regardless of the law.
Enforced by courts, fines, penalties.	Enforced by conscience, reputation, public pressure.
The same everywhere the law applies.	Can differ between people, cultures and contexts.
Emerging tech	New ethical risk it creates
AI-generated phishing	Scam emails now sound perfect — harder for anyone to spot, so the harm spreads wider.
Deepfakes	A faked video or voice can impersonate a real person, destroying trust and reputations.
IoT / smart devices	Cheap sensors with poor security become easy entry points into a whole network.
Cloud storage	One breach can expose <i>millions</i> of records at once, not one filing cabinet.
Personalisation people enjoy	Profiling and surveillance creep beyond the original purpose
Physiological (who you are)	Behavioural (how you act)
Business owner	Lower costs, fewer errors, 24/7 service — clear efficiency gains.
Displaced worker	Lost income and identity; “reskill” is hard if you're 55 or rural.
Customer	Cheaper, faster service — but misses the human touch and help.
Society / government	New industries and jobs created, but rising inequality and welfare cost if transition isn't managed.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Identify · 2 marks

— 1 — definition: data about a physiological/behavioural characteristic used to identify a person. 1 — a valid example (fingerprint, face, iris, voice, gait).

✗ A weak answer

Biometric data is your personal data like your password and login. A password is not biometric — this confuses a chosen secret with a body measurement. No valid example, so 0 marks.

✓ A strong answer

Biometric data is information about a person's physical or behavioural characteristics used to identify them. An example is a fingerprint (or facial scan). Clear definition (1) + a correct, specific example (1). Full marks.

Q2. Practice question · 3 marks

— 1 — a password can be reset/changed after a breach. 1 — a biometric is permanent and cannot be changed. 1 — consequence: lifelong exposure / reuse for fraud / harm to the member.

✗ A weak answer

It's worse because faces are private and people would be upset if they were leaked. True but shallow — "people would be upset" applies to any breach. Misses the actual reason: permanence. Likely 1 mark.

✓ A strong answer

A password is a secret that can be changed: if it leaks, a member simply resets it and the threat ends. A facial scan is a permanent feature of a person's body that cannot be changed. If PulsePoint's biometric data is breached, the exposure is lifelong — the member can never "reset" their face — and it could be reused for identity fraud or false matches anywhere facial recognition is used. Names the key contrast (resettable vs permanent), explains the consequence (lifelong exposure, reuse), and links to PulsePoint members. Full marks.

Q3. Practice question · 4 marks

— 1 — identifies job displacement as the harm (lost income/role). 1 — gives a benefit (lower cost / cheaper access / new roles). 1 — names specific stakeholders affected (staff, customers, business). 1 — reasoned judgement, e.g. fairness depends on managing the transition (retraining/notice).

✗ A weak answer

Replacing staff with machines is bad because people lose their jobs and robots are taking over. PulsePoint shouldn't do it. One-sided and emotive. No benefit considered, no specific stakeholders, no real "discussion". Around 1 mark.

✓ A strong answer

The automation creates job displacement: PulsePoint's receptionists lose their income and the role they identify with, which is the main ethical harm. However, there is a genuine benefit — lower running costs let PulsePoint offer cheaper memberships and 24/7 access, and it may create new roles such as kiosk technicians. The ethical concern is therefore less about *whether* to automate and more about *how*: dismissing staff with no notice, retraining or redeployment would be unethical, whereas offering reskilling support would treat them fairly. Customers also lose the human help some members rely on. On balance, automation is defensible only if PulsePoint manages a fair transition for displaced workers. Both sides (harm + benefit), specific stakeholders (staff, business, customers), the "how not whether" insight, and a reasoned conclusion. Full marks.

Q4. Practice question · 4 marks

— 1 — legal = follows the law; ethical = fairness/right vs wrong. 1 — states they are distinct (legal ≠ automatically ethical). 1 — applies to PulsePoint (e.g. consent isn't meaningful if you can't refuse). 1 — identifies the remaining ethical concern (permanence / misuse / understanding).

✗ A weak answer

It's legal because they told people with a sign, so it follows the Privacy Act and is fine. Answers the legal question only and concludes "fine" — the opposite of what's asked. Misses the ethical dimension entirely. ~1 mark.

✓ A strong answer

Legality means an action follows the law; ethics asks whether it is fair and respectful of those affected — they are not the same. PulsePoint's sign may make collection *legal*, but it can still be *unethical*: a teenager who wants to train with friends can't realistically refuse, so their "consent" isn't freely given. The data is also permanent and could later be misused, and members may not understand what they've agreed to. So even fully within the law, PulsePoint raises ethical concerns about meaningful consent and the long-term risk to members. Defines the distinction, applies it to PulsePoint with a concrete consent example, and explains the residual ethical concern. Full marks.

Revision checklist

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- The dot point: ethical issues of emerging tech — cyber threats, biometric data, job displacement. All three examinable.
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